

Q1. Given N array element, choose 2 indices i, j , such that $i \neq j$ and $[arr[i] \& arr[j]]$ is max^m, return the max^m value of $arr[i] \& arr[j]$.
 $\rightarrow$ bitwise &

ex $\Rightarrow arr[2] \Rightarrow \{27, 18, 20\} \rightarrow O(p) [27 \& 18] \rightarrow \underline{18}$

$$\begin{array}{r} 27 \& 18 \\ 27: \quad 11011 \\ 18: \quad 10010 \\ \hline 10010 \\ \hline \downarrow \\ 18 \end{array}$$

$$\begin{array}{r} 27 \& 20 \\ 27: \quad 11011 \\ 20: \quad 10100 \\ \hline 10000 \\ \hline \downarrow \\ 16 \end{array}$$

$$\begin{array}{r} 18 \& 20 \\ 18: \quad 10010 \\ 20: \quad 10100 \\ \hline 10000 \\ \hline \downarrow \\ 16 \end{array}$$

ex $\Rightarrow arr[4] \Rightarrow \{21, 18, 24, 17\}$

$\rightarrow O(p) [21 \& 17] = 17$

Bruteforce

[check for all possible pairs]

iterations $\Rightarrow \frac{n(n+1)}{2}$

$TC \Rightarrow O(N^2)$ $SC \Rightarrow O(1)$
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Optimise

→ use sorting heap us → [x]

ex: arr[7] = { 26, 13, 23, 28, 27, 7, 25 }

		4	3	2	1	0	
26 →	1	1	0	1	0		
13 →	0	0	0	0	0	-4	
23 →	1	0	0	0	0	-3	
28 →	1	1	1	0	0		
27 →	1	1	0	1	1		
7 →	0	0	0	0	0	-4	
25 →	1	1	0	0	1		
	1	1	0	1	0		→ 26

26 & 27

Pseudo

int ans = 0

for (i=31; i>=0; i--) {

 c = 0;

 for (j=0; j<N; j++) {

 if (checkBit(arr[j], i)) {

 c++

 }

 if (c >= 2) {

 ans = ans + (1 << i)

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        for (j=0; j<N; j++) {
            if (!checkbit(arr[j], i)) {
                arr[j] = 0
            }
        }
    }
    return

```

TC $\Rightarrow (32 \times 2N)$
 $\Rightarrow 64N \approx O(N)$
 SC $\Rightarrow O(1)$

Inverse Modulo

Given, a, m and b $\rightarrow$ b is inverse modulo of a.
 if,

$$[(a \times b) \% m = 1]$$

* b exists if $\text{gcd}(a, m) = 1$

* value of b in range $[2, m-1]$.

ex $\Rightarrow$ a = 10, m = 7, check & find Inv. modulo?

$\Rightarrow \text{gcd}(a, m) \Rightarrow \text{gcd}(10, 7) = 1$ ✓

$$11) b \in [2, 6]$$

$$b=2, \quad (a \times b) \% m = (10 \times 2) \% 7 = 6$$

$$b=3 \quad (10 \times 3) \% 7 = 2$$

$$b=4 \quad (10 \times 4) \% 7 = 5$$

$$b=5 \quad (10 \times 5) \% 7 = 1$$

$$(10 \times 5) \% 7 = 1$$

$$b=5 \rightarrow \text{Inv. modulo}$$

ex 3 $a=12, m=5$, check and find inv modulo

$$1) \gcd(a, m) = \gcd(12, 5) = 1 \quad \checkmark$$

$$11) b \in [2, 4]$$

$$b=2 \rightarrow (a \times b) \% m = (12 \times 2) \% 5 = 4$$

$$b=3 \quad (12 \times 3) \% 5 = 1$$

$$\underline{\underline{b=3}}$$

Q Given a, m find the inv modulo of a , if exists?

$$\textcircled{1} \quad \gcd(a, m) = 1$$

$$\text{if } (\gcd == 1) \{$$

$$\text{for } (i=2; i < m; i++) \{$$

$$\textcircled{11}$$

$$\text{if } [(a \times i) \% m == 1]$$

$$\left[\begin{array}{l} TC = O(M) \\ SC = O(1) \end{array} \right]$$

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    }
    return i
}
return t

```

Special case

→ p is prime no.

→ $a \% p \neq 0$

↓

a is not a multiple of p

$$\left[\begin{array}{l} \text{i) } a^{-1} \% p = (a^{p-2}) \% p \\ \text{ii) } (a^{p-1} \% p) = 1 \end{array} \right] \text{fermat little theorem}$$

Q Max non negative subarray

Given $arr[N]$, find max sum subarray of non-negative nos. of A .

ex $\Rightarrow A = [1, 2, 5, -7, 2, 3]$

$\downarrow$
 8

o/p $\Rightarrow [1, 2, 5]$

$A = [10, -1, 2, 3, -4, 100]$

$\downarrow$ $\downarrow$ $\downarrow$
 10 5 100

o/p $\Rightarrow [100]$

$$A = [-10 \ 10 \ 6 \ -2 \ 50 \ -3 \ -4 \ 99]$$

$$O/p \Rightarrow [99]$$

$$[-10 \ 10 \ 6 \ -2 \ 50 \ -3 \ -4 \ 99]$$

$$\Rightarrow \text{ans} = \begin{bmatrix} 99 \\ \cancel{50} \end{bmatrix} \quad \text{ansken } 99$$

$$\Rightarrow \text{current} = [99] \quad \text{current } 99$$

$$[99 \ 64 \ 18 \ -2 \ 101 \ 62 \ 13 \ -3 \ -4 \ 100]$$

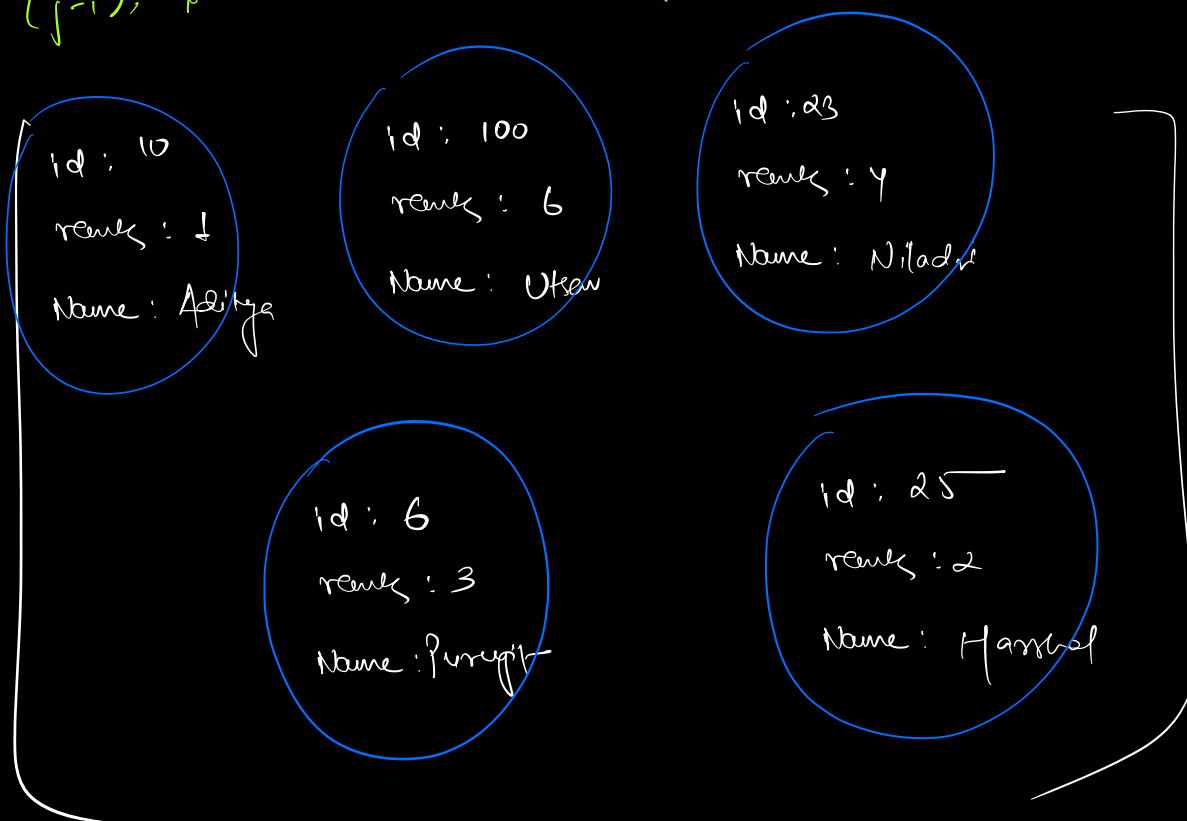
$$\text{ansArr} = \underline{[99, 64, 18]} \quad \text{ansken } 181$$

$$\text{current} = [100] \quad \text{current } 100$$

$$\boxed{\begin{array}{l} \text{TC} \Rightarrow O(N) \\ \text{SC} \Rightarrow O(N) \end{array}}$$

Q Max distance!

Given an array A of size N. find the max value of $(j-i)$, such that $A[i] \leq A[j]$



⇒ Sort by creating a comparator $\rightarrow$ $\left. \begin{array}{l} \text{id} \\ \text{rank} \\ \text{name} \end{array} \right\}$

ex ⇒ arr = $\begin{array}{cccc} 0 & 1 & 2 & 3 \\ [3 & 5 & 4 & 2] \end{array}$

$\underbrace{\hspace{1.5cm}}_i \quad \underbrace{\hspace{2.5cm}}_j$

$\left. \begin{array}{l} i \Rightarrow 0 \quad A[i] = 3 \\ j \Rightarrow 1 \quad A[j] = 5 > 3 \\ \quad 2 \quad A[j] = 4 > 3 \end{array} \right\}$

$\text{max}(j-i)$
 $= 2 - 0 = \underline{\underline{2}}$

Brute force

check all possible pairs for (i, j)

$$\begin{array}{l} TC = O(N^2) \\ SC = O(1) \end{array}$$

Optimize

arr $\Rightarrow$ $\begin{array}{cccc} 0 & 1 & 2 & 3 \\ 3 & 5 & 4 & 2 \end{array}$

$\rightarrow$ direct sorting will not help
because it will shuffle the index
& element

pos/arr $\Rightarrow$ $\begin{bmatrix} (3, 0) & (5, 1) & (4, 2) & (2, 3) \end{bmatrix}$

Sort this
arr on the
basis of val
in asc

$\rightarrow$ $\begin{bmatrix} (2, 3) & (3, 0) & (4, 2) & (5, 1) \end{bmatrix}$

Can be done
using comparator

max1 $\rightarrow$ ~~4~~ 2

diff $\rightarrow$ ~~4~~ 1

maxDiff $\rightarrow$ $\boxed{2}$ $\rightarrow$ $\frac{0}{1}$

$[i, j]$ $\rightarrow$ max m
 $\downarrow$ $\downarrow$
 max m min m

$TC \rightarrow O(N \log N)$ $SC \rightarrow O(N)$
